

Investigation of d - and q -Axis Inductances Influenced by Slot-Pole Combinations Based on Axial Flux Permanent-Magnet Machines

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Abstract—Flux-weakening capability of permanent-magnet synchronous machines (PMSMs) is fundamentally determined by d - and q -axis inductances. Based on axial flux permanent-magnet machines (AFPMs), this paper extracts the intrinsic factors that influence the d - and q -axis inductances of PMSMs, meanwhile revealing the fact that the d - and q -axis inductances of fractional slot concentrated winding machines are not inevitably larger than that of distributed ones. The proportion of each of the d - and q -axis component inductances for multigap AFPMs is first studied. By analyzing armature inductance in conjunction with differential leakage inductance, this paper introduces a novel parameter to decouple the influence of slot-pole combinations and physical dimensions so that the “inductance ability” of different slot-pole combinations can be quantified. An improved method of calculating the slot leakage inductance of semiclosed slots is then put forward, which proves to be more accurate than the conventional method. The investigations and conclusion in this paper are not restricted to AFPMs but applicable to radial flux permanent-magnet machines as well. Validations of various slot-pole combinations are carried out with finite-element analysis and experiments.

Index Terms—Axial flux permanent-magnet machine (AFPM), d - and q -axis inductances, flux weakening, fractional slot concentrated winding, slot leakage inductance.

NOMENCLATURE

$L_{d,q}$	d - and q -axis inductances.
K_{ms}	Coefficient related to the type of axial flux permanent-magnet machines (AFPMs).
$L_{gd,gq}$	d - and q -axis air gap inductances for a single air gap.
$L_{ad,aq}$	d - and q -axis armature inductances for a single air gap.
$L_{\delta d,\delta q}$	d - and q -axis differential leakage inductances for a single air gap.
L_s	Slot leakage inductance for a single air gap.
L_t	Tooth tip leakage inductance for a single air gap.
L_e	End leakage inductance for a single air gap.
λ	Specific permeance.
m	Number of phases (in this paper, $m = 3$).
p	Number of pole pairs.

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Q	Number of slots for a single air gap.
q	Slots per pole per phase (SPP) for a single air gap.
N_1	Number of turns per phase for a single air gap.
D_{equ}	Equivalent diameter for analytical calculation.
l_{ef}	Effective length of the stator core.
δ_1	d -axis air gap length.
δ_2	q -axis air gap length excluding δ_1 .
v	Order of the armature v th harmonic.
K_{dpv}	Winding factor of the armature v th harmonic.
Σg	Coefficient of air gap specific permeance.
$\chi_{gd,gq}$	Coefficient related to physical dimensions for air gap inductance.
y_1	Coil pitch.

I. INTRODUCTION

MODERN ELECTRIC drive systems are claiming higher demands on permanent-magnet synchronous machines (PMSMs) in power density and constant power speed range (CPSR). As one of the most concerned issues, the flux-weakening capability of PMSMs has been widely studied and has been found to be determined by the characteristic current [1]–[3]

$$I_{\text{ch}} = \Psi_f / L_d \quad (1)$$

where Ψ_f is the permanent-magnet flux linkage. A relatively large L_d is beneficial to the reduction in d -axis current supplied by the inverters for flux weakening; hence, it saves the capacity of driving systems. Therefore, it is crucial to design L_d suitable enough to deduce a good output performance [4].

Previous work believed that interior permanent-magnet machines (IPMs) were better candidates for achieving wide CPSR under flux-weakening operation, whereas El-Refaie *et al.* found that surface-mounted permanent-magnet machines (SPMs) with fractional slot concentrated windings could also offer large d -axis inductances and hence could output a wide CPSR [5]–[7]. Furthermore, they pointed out that fractional slot concentrated winding machines took advantage in d -axis inductances compared with distributed ones because of their large harmonic leakage inductances and considered SPP values of 2/5 or 2/7 as the optimal slot-pole combinations. However, quantitative explanations were not available in their works. In fact, it will be demonstrated in this paper that the d -axis

inductances of the former are not inevitably larger than that of the latter, and optimal combinations are not 2/5 or 2/7 neither.

The d - and q -axis inductances consist of armature inductance, differential leakage inductance, slot leakage inductance, tooth tip leakage inductance, and end leakage inductance, which are commonly adopted in analytical calculations. As the end leakage inductance is greatly affected by manufacture and much smaller than the others, it will not be discussed in this paper. Instead, this paper focuses on the intrinsic factors of how d - and q -axis inductances are influenced by slot-pole combinations. For detailed works on end leakage inductance, please refer to [8]–[11].

Previous work tended to analyze and calculate the armature inductance and differential leakage inductance separately because of their different contributions to the air gap flux. However, when output capability is the major concern, there is no discrimination between them. In fact, armature magnetic field and differential leakage magnetic field are fundamentally similar in the view that they are both generated by armature excitation and directly influenced by slot-pole combinations. This paper discusses and calculates them in the same section and defines the united inductance as air gap inductance, which is beneficial in explaining the effects of slot-pole combinations.

Slot leakage inductance is also important especially when semiclosed slots are adopted for high performance. Conventional analytical methods assumed the slot leakage flux curve parallel to the slot bottom, which was accurate enough for open slots [12]–[14]. However, when it comes to semiclosed slots, such assumption will cause inevitable error. Considering the real leakage flux distribution in the slot, this paper introduces an improved method for both single-layer and double-layer semiclosed slots, which is proved to be more accurate. Influence of core saturation is also illustrated for supplementation.

All of the d - and q -axis component inductances are investigated based on AFPMs which are competitive in wheel drives, ship propulsions, wind turbines, etc., because of their compactness, shape flexibility, and high torque density [15]–[21]. In particular, when the axial length is too small for radial flux permanent-magnet machine (RFPM) design, AFPMs are undoubtedly better choices because their output power has little relationship with axial length. In fact, investigations in this paper are also applicable to RFPMs and are demonstrated via finite-element analysis (FEA) and experiments.

Before the study of component inductances, it is essential to consider the proportions that they contribute to d - and q -axis inductances in multigap AFPMs which were seldom published [22]–[24]. Based on dual-gap AFPMs, this paper first proposes d - and q -axis inductance expressions in Section II, which can also be extended to multigap cases. Second, comprehensive flux distribution in AFPMs is analyzed to determine the equivalent circumferential section for inductance calculation. Then, in Section IV, d - and q -axis component inductances are investigated with the proposed method, and laws on how d - and q -axis inductances are influenced by slot-pole combinations are revealed. Finally, the d - and q -axis inductances of various slot-pole combinations of AFPMs and RFPMs are compared with analytical approach and FEA, and the proposed method is further verified by experiments.

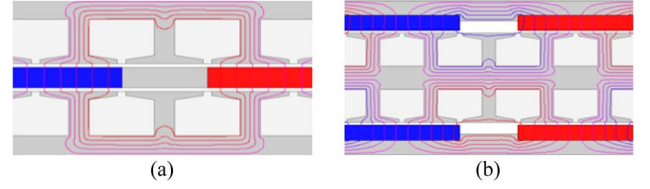


Fig. 1. Circumferential flux distribution of typical dual-gap AFPMs. (a) AFIR. (b) TORUS-NN.

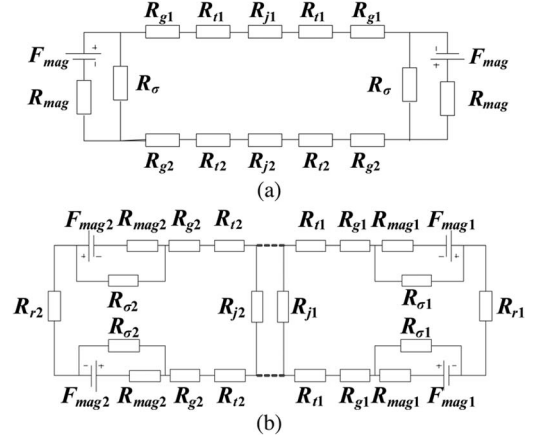


Fig. 2. Equivalent magnetic circuits of AFPMs. (a) Series equivalent magnetic circuit. (b) Parallel equivalent magnetic circuit.

II. INDUCTANCE COMPOSITION OF MULTIGAP AFPMS

Multigap AFPMs can be basically classified into two types according to their equivalent magnetic circuits, i.e., series and parallel. The circumferential view of two typical dual-gap AFPM structures is shown in Fig. 1, where Fig. 1(a) belongs to the series type (the flux curve of one pole cuts through all of the gaps and cores) and Fig. 1(b) belongs to the parallel type (the flux curve of one pole cuts through only part of the machine).

The equivalent series and parallel magnetic circuits of Fig. 1 are illustrated in Fig. 2(a) and (b), respectively, where F_{mag} is the permanent-magnet MMF, R_{mag} is the permanent-magnet reluctance, R_{σ} is the leakage reluctance, R_g is the air gap reluctance, R_t is the stator tooth reluctance, R_j is the stator yoke reluctance, and R_r is the rotor core reluctance. The subscripts 1 and 2 represent the two air gaps, respectively.

From the equivalent magnetic circuits, the evaluation of the d - and q -axis inductances for dual-gap AFPMs is proposed as

$$L_{d,q} = 2(K_{ms}L_{gd,q} + L_s + L_t + L_e) \quad (2)$$

where $L_{gd,q}$ are the d - and q -axis air gap inductances for a single air gap consisting of d - and q -axis armature inductances and d - and q -axis differential leakage inductances, which will be expounded in Section IV; L_s is the slot leakage inductance per phase for a single air gap; L_t is the tooth tip leakage inductance for a single air gap; and L_e is the end leakage inductance for a single air gap. K_{ms} depends on the magnetic circuit type of AFPMs. For the series type, $K_{ms} = 2$; for the parallel type, $K_{ms} = 1$.

In fact, the coefficient K_{ms} reflects the coupling of armature magnetic field between two air gaps. For the series type, the armature reactance field of two air gaps fully couples, which

means that the air gap mutual inductance between two stator windings is equal to the air gap self-inductance of one stator; hence, $K_{ms} = 2$. For the parallel type, there is no coupling between two stator armature windings; hence, $K_{ms} = 1$. For the other leakage inductances, there is hardly any coupling between two air gaps so that their coefficients are always unity.

The conclusion can also be demonstrated via 3-D FEA. Take an AFIR prototype with 30 slots and 10 poles for example. Based on the definition of inductance

$$L = \Delta\psi / \Delta i \quad (3)$$

this paper calculated that the self-inductance with single stator excited was 2.31 mH. When both stator windings were excited, this paper obtained the *d*-axis inductance as 6.38 mH. Obviously, the latter is neither twice nor four times of the former, inferring that the contribution of each component inductance is different in dual-gap AFPMs. Similar derivations can also be applied to multigap AFPMs.

For analytical calculations, inductance can be expressed in terms of specific permeance as

$$L_x = 2\mu_0 \frac{N_1^2}{pq} l_{ef} \lambda_x \quad (4)$$

where λ_x is the specific permeance, N_1 is the number of series turns per phase for a single air gap, p is the number of pole pairs, q is the SPP for a single air gap, μ_0 is the magnetic permeability of free space, and l_{ef} is the effective length of the stator core, which is

$$l_{ef} = K_{Fe}(D_o - D_i)/2 \quad (5)$$

where D_o and D_i are the outer and inner diameters of the AFPM air gap, respectively, and K_{Fe} is the stacking factor of laminations.

Hence, the calculation of inductance can be simplified as the calculation of related specific permeance. Before that, it is essential to have a view of the magnetic flux distribution in AFPMs in order to determine the equivalent section for specific permeance calculation.

III. SPATIAL DISTRIBUTION OF MAGNETIC FIELD IN AFPMs

In RFPMs, the diameter of the armature core is generally adopted as the equivalent diameter for inductance calculations, which is different for AFPMs because of their unique structures and complicated flux distribution. In this section, a 3-D AFIR model with 36 slots and 40 poles is built for observation in order to determine the equivalent diameter for analytical evaluations. Other major parameters are the following: core length of 70 mm, stator height of 14 mm, yoke height of 3 mm, and trapezoidal magnet.

A. Air Gap Flux Distribution in AFPMs

The air gap flux distribution in AFPMs sweeping 360 electrical degrees is shown in Fig. 3(b), and the radius coordinate starts from the inner side of the AFPM stator. It can be seen

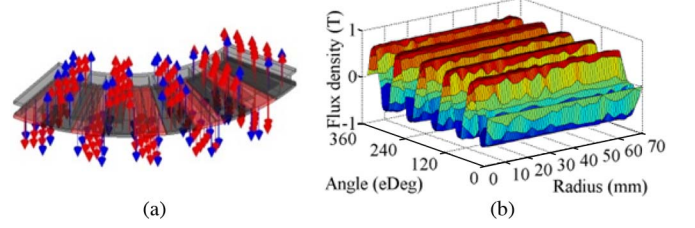


Fig. 3. Distribution of air gap flux density. (a) Flux density vector on the sampling section. (b) Distribution of the axial component of the air gap flux density.

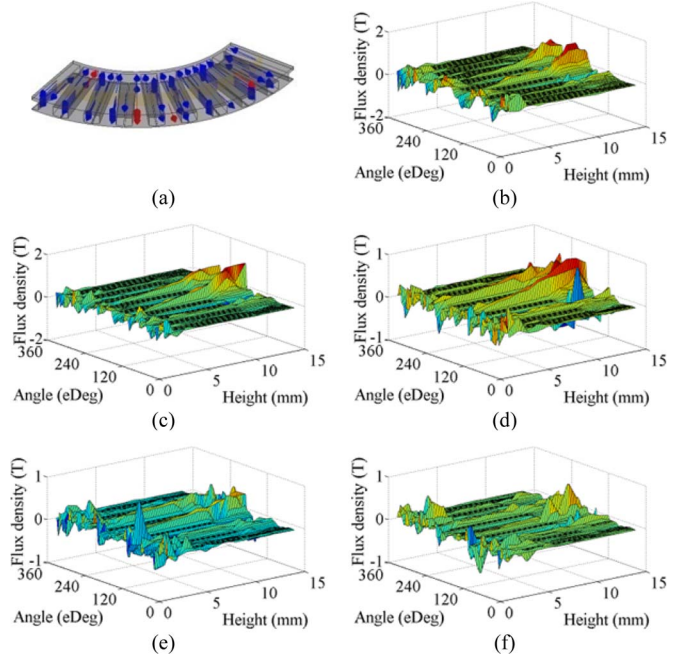


Fig. 4. Radial component of the flux density at different radii in the AFPM stator. (a) Flux density vector on the sampling section. (b) $\Delta R = 10$ mm. (c) $\Delta R = 25$ mm. (d) $\Delta R = 35$ mm. (e) $\Delta R = 45$ mm. (f) $\Delta R = 60$ mm.

that the flux density along the radial direction varies a little. Therefore, in the view of the air gap, any section along the circumference can be adopted for calculation.

B. Stator Flux Distribution in AFPMs

Distinct from RFPMs, the flux distribution in AFPM stators is typically in three dimensions, which contains not only axial and circumferential but also radial components. The former two contribute to the permanent-magnet flux linkage coupled with armature windings, while the last only aggravates the saturation and wastes the permanent-magnet flux. The radial components of the flux density sweeping 360 electrical degrees at different radii offset from the inner side of the stator, i.e., $\Delta R = 10, 25, 35, 45$, and 60 mm, are illustrated in Fig. 4(b)–(f), respectively. It can be seen that the radial component is larger at the tooth tip and yoke bottom and decreases along the radial direction.

The axial component of the flux density on the planer section in the tooth sweeping 360 electrical degrees is shown in Fig. 5(b), and the circumferential component of the flux density

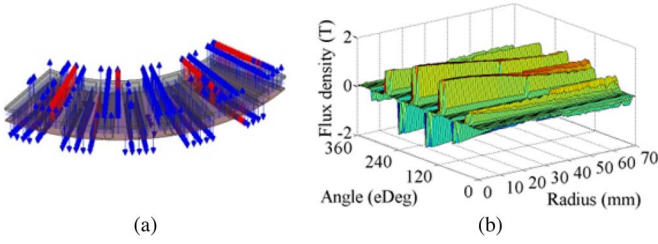


Fig. 5. Axial component of the flux density in the AFPM stator tooth. (a) Flux density vector on the sampling section. (b) Flux density distribution.

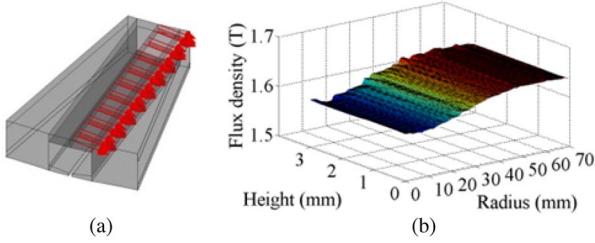


Fig. 6. Circumferential component of the flux density in the AFPM stator yoke. (a) Flux density vector on the sampling section. (b) Flux density distribution.

on the planer section in the stator yoke is shown in Fig. 6(b). It can be seen that both two components decrease along the radial direction and are approximately odd symmetric about the average diameter.

From the deduced observations, it is reasonable to estimate the equivalent electromagnetic characteristics on the circumferential section at the average diameter of the AFPMs in view of the equivalence of the permanent-magnet flux. Hence, the equivalent diameter of the AFPMs for evaluation can be regarded as

$$D_{\text{equ}} = (D_o + D_i)/2. \quad (6)$$

IV. INVESTIGATION OF COMPONENT INDUCTANCES

Based on previous observations, the circumferential section at the average diameter of the AFPMs is adopted to evaluate the component inductances in order to obtain the d - and q -axis inductances. The derivations in this section are also applicable to RFPMs and are validated later.

A. Air Gap Inductance

By evaluating armature inductance in conjunction with differential leakage inductance, this paper defines the sum as air gap inductance, which is

$$L_{gd,qq} = L_{ad,aq} + L_{\delta d,\delta q} \quad (7)$$

where $L_{ad,aq}$ are the d - and q -axis armature inductances and $L_{\delta d,\delta q}$ are the d - and q -axis differential leakage inductances.

The d - and q -axis armature inductances for a single air gap can be derived as

$$L_{ad,aq} = \frac{m}{\pi} D_{\text{equ}} l_{\text{ef}} \frac{(N_1 K_{dp1})^2}{p^2} \frac{K_{d,q}}{K_{sd,sq}} \lambda_{ad,aq} \quad (8)$$

where $K_{d,q}$ are armature reaction factors in d - and q -axes; for SPM, $K_d = K_q = 1$. $K_{sd,sq}$ are the saturation factors of the d - and q -axis magnetic circuits and are assumed to be equal to unity in this paper. $\lambda_{ad,aq}$ are the specific permeances of the d - and q -axis magnetic circuits and can be written as

$$\begin{cases} \lambda_{ad} = \mu_0 / (K_{ms} K_{\delta} \delta_1 + h_m / \mu_r) \\ \lambda_{aq} = \mu_0 / (K_{ms} K_{\delta} \delta_1 + \delta_2) \end{cases} \quad (9)$$

where h_m is the magnet thickness, δ_1 is the d -axis air gap length, δ_2 is the q -axis air gap length excluding δ_1 (for SPM, $\delta_2 = h_m$), K_{δ} is the Carter's coefficient at D_{equ} , and μ_r is the relative magnetic permeability of the magnet.

The formula of the harmonic leakage inductance is similar. The only difference is to substitute the winding factor and pole pair of v th order instead of fundamental, i.e.,

$$L_{vd,vq} = \frac{m}{\pi} D_{\text{equ}} l_{\text{ef}} \frac{(N_1 K_{dpv})^2}{(vp)^2} \frac{K_{d,q}}{K_{sd,sq}} \lambda_{ad,aq} \quad (10)$$

where v is the number of the air gap v th harmonic; for fundamental, $v = 1$. K_{dpv} is the winding factor of the v th harmonic.

Combining (8) and (10) and defining a novel parameter named coefficient of air gap specific permeance as

$$\Sigma g = \sum [K_{dpv} / (vp)]^2 \quad (11)$$

this paper deduces the d - and q -axis air gap inductances as

$$L_{gd,qq} = \frac{m}{\pi} D_{\text{equ}} l_{\text{ef}} (N_1^2 \Sigma g) \frac{K_{d,q}}{K_{sd,sq}} \lambda_{ad,aq}. \quad (12)$$

Furthermore

$$L_{gd,qq} = \chi_{gd,qq} \cdot \Sigma g \quad (13)$$

where Σg only relates to slot-pole combination and $\chi_{gd,qq}$ relate to physical dimensions and saturation

$$\chi_{gd,qq} = \frac{m}{\pi} D_{\text{equ}} l_{\text{ef}} N_1^2 \frac{K_{d,q}}{K_{sd,sq}} \lambda_{ad,aq}. \quad (14)$$

The formula of Σg is presented in the Appendix, which is universal for both fractional and integral slot windings. Hence, the air gap inductance is decoupled to a production of physical dimensions and slot-pole combinations.

Figs. 7 and 8 illustrate the relationship between Σg and the ratio of coil pitch to pole pitch β , with the pole and slot number invariant, respectively, where

$$\beta = y_1 / (mq) \quad (15)$$

and y_1 is the coil pitch.

As the numbers of pole pairs for all of the curves in Fig. 7 are the same, they can be normalized to unity. The slot numbers in Fig. 8 are all 36. Two curves coincide with each other if their SPP values satisfy

$$1/q_1 + 1/q_2 = 2m. \quad (16)$$

It can be seen from Fig. 7 that fractional slot concentrated winding machines can offer larger air gap inductances than distributed ones with the same pole number, which proves

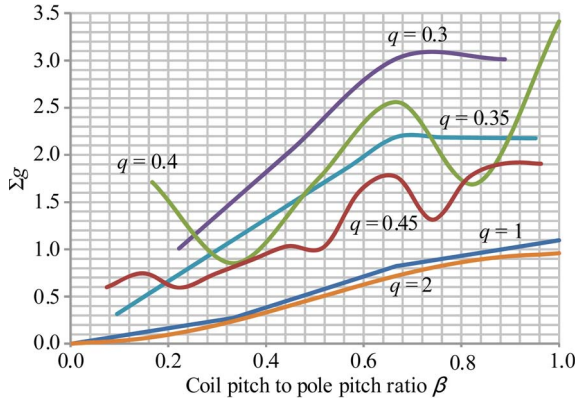


Fig. 7. Coefficient of air gap specific permeance Σg with respect to the coil pitch to pole pitch ratio β of different slot-pole combinations when the pole number remains invariant.

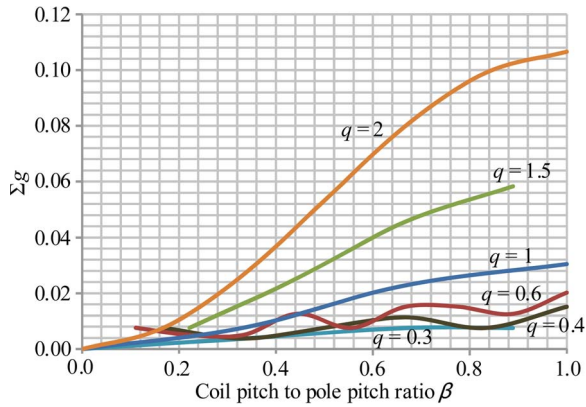


Fig. 8. Coefficient of air gap specific permeance Σg with respect to the coil pitch to pole pitch ratio β of different slot-pole combinations when the slot number remains invariant.

El-Refaie's work. However, it can also be seen that $q = 2/5$ or $2/7$ are not the best combinations to produce the largest air gap inductance for the same $\chi_{gd,qq}$ considering $q = 3/10$ or $3/8$. Furthermore, if the slot number is invariant as shown in Fig. 8, fractional slot concentrated winding machines lose out to distributed ones.

The coefficient of air gap specific permeance Σg of numerous slot-pole combinations is calculated and listed in Table I. Coil pitch y_1 is determined to achieve the largest fundamental winding factor k_{dp1} . The bold frames distinguish the concentrated windings from distributed ones. For combinations with the same SPP values, the coil pitch and fundamental winding factor are omitted because they are also the same. However, Σg may be not because of different pole numbers. Therefore, only q and Σg are presented for the same SPP. The negative k_{dp1} means that its magnetic field rotates in the negative direction compared with the positive one. From Table I, several inferences can be drawn as follows.

- 1) Concentrated windings occur in a period for the same slot number, i.e., the row of the table.
- 2) For the same slot number in the row, Σg values are the same if their SPP values satisfy (16).

- 3) Generally, for the same pole number in the column, the smaller the slot number is, the larger the Σg is, except for some cases where the concentrated winding appears in the middle way, which further infers the following.
- 4) Even when pole numbers are the same, concentrated windings may not offer larger air gap inductances than distributed ones.

This table is helpful for designers to choose the proper slot-pole combinations in the view of inductance. For example, although the combinations of 4 poles and 12 slots, and 8 poles and 24 slots share the same q , it can be seen from the table that Σg of the former is much larger than that of the latter, which means that the former may produce a larger air gap inductance with the same $\chi_{gd,qq}$. A larger inductance contributes to a wider CPSR but may reduce the maximum output power. Therefore, choosing a suitable Σg is crucial for a proper design.

It should be noted that d - and q -axis inductances may not be exactly proportional to Σg because of the other leakage inductances and difference in $\chi_{gd,qq}$. However, as the air gap inductances usually dominate the d - and q -axis inductances, the "inductance ability" of different slot-pole combinations can still be estimated with the help of (13).

B. Slot Leakage Inductance

Slot leakage inductance is sometimes very large especially when semiclosed slots are adopted for high performance. Both FEA and analytical method can be used to calculate slot leakage inductance.

When FEA is applied to calculate slot leakage inductance, two assumptions are made in advance.

- 1) The permeability of the stator core is large enough so that the magnetic curve cuts through slot edges perpendicularly.
- 2) The magnetic curve on the slot opening edge is parallel to the slot bottom.

Then, the magnetic vector potential in the slot satisfies

$$\begin{cases} \nabla^2 A = -\mu J \\ \partial A / \partial n = 0 \\ A_0 = \text{const} \end{cases} \quad (17)$$

where A_0 is the magnetic vector potential on the slot opening.

Then, the energy produced by the leakage magnetic field in the slot can be deduced by integration as

$$W_{ss} = \frac{1}{2\mu_0} l_{ef} \int_s B^2 ds \quad (18)$$

and the slot leakage inductance can be obtained as

$$L_{ss} = 2W_{ss}/I^2. \quad (19)$$

Furthermore, the slot leakage inductance can be written in the form of specific permeance as

$$L_{ss} = N_s^2 \mu_0 l_{ef} \lambda_{ss} \quad (20)$$

TABLE I
 Σg OF DIFFERENT SLOT-POLE COMBINATIONS

Slot number		Pole number							
		4	8	10	14	16	20	22	26
6	q	0.5	0.25	0.2	0.143	0.125	0.1	0.091	0.077
	y_1	1	1	3	3	1	1	3	3
	k_{dp1}	0.866	0.866	1	-1	-0.866	-0.866	-1	1
	Σg	0.273	0.273	1.096	1.096	0.273	0.273	1.096	1.096
9	q	0.75	0.375	0.3	0.214	0.188	0.15	0.136	0.115
	y_1	2	1	1	2	4	4	2	1
	k_{dp1}	0.945	0.945	0.945	-0.945	-0.945	0.945	0.945	-0.945
	Σg	0.283	0.121	0.121	0.283	0.933	0.933	0.283	0.121
12	q	1		0.4	0.286			0.182	0.154
	y_1	3	0.5	1	1	0.25	0.2	5	5
	k_{dp1}	1		0.933	0.933			0.933	-0.933
	Σg	0.274	0.068	0.068	0.068	0.068	0.205	0.891	0.891
15	q	1.25	0.625		0.357	0.313		0.227	0.192
	y_1	4	2	0.5	1	1	0.25	2	4
	k_{dp1}	0.951	0.951		0.951	0.951		-0.951	-0.951
	Σg	0.254	0.078	0.044	0.043	0.043	0.044	0.078	0.254
18	q	1.5		0.6	0.429	0.375		0.273	0.231
	y_1	4	0.75	2	1	1	0.3	1	2
	k_{dp1}	0.945		0.945	0.902	0.945		0.902	-0.945
	Σg	0.233	0.071	0.050	0.030	0.030	0.030	0.030	0.050
21	q	1.75	0.875	0.7		0.438	0.35	0.318	0.269
	y_1	5	3	2	0.5	1	1	1	1
	k_{dp1}	0.953	0.932	0.953		0.89	0.953	0.953	0.89
	Σg	0.239	0.070	0.047	0.022	0.022	0.022	0.022	0.022
24	q	2		0.8	0.571			0.364	0.308
	y_1	6	1	2	2	0.5	0.4	1	1
	k_{dp1}	0.966		0.925	0.925			0.949	0.949
	Σg	0.240	0.068	0.043	0.025	0.017	0.017	0.017	0.017
27	q	2.25	1.125	0.9	0.643	0.563	0.45	0.409	0.346
	y_1	7	4	3	2	2	1	1	1
	k_{dp1}	0.954	0.915	0.941	0.954	0.915	0.877	0.915	0.954
	Σg	0.236	0.061	0.045	0.025	0.019	0.013	0.013	0.013
30	q	2.5			0.714			0.455	0.385
	y_1	8	1.25	1	2	0.625	0.5	1	1
	k_{dp1}	0.951			0.951			0.874	0.936
	Σg	0.230	0.063	0.044	0.024	0.020	0.011	0.011	0.011
33	q	2.75	1.375	1.1	0.786	0.688	0.55		0.423
	y_1	8	4	3	2	2	2	0.5	1
	k_{dp1}	0.954	0.954	0.946	0.928	0.954	0.903		0.903
	Σg	0.233	0.061	0.041	0.022	0.019	0.012	9.036e ⁻³	8.719e ⁻³

where N_s is the number of conductors in the slot. From (18)–(20), the slot leakage specific permeance can be obtained as

$$\lambda_{ss} = \frac{L_{ss}}{N_s^2 \mu_0 l_{ef}} = \frac{2W_{ss}}{I^2 N_s^2 \mu_0 l_{ef}} = \frac{\int_s B^2 ds}{I^2 N_s^2 \mu_0^2}. \quad (21)$$

When the conventional analytical method is applied to calculate λ_{ss} , another assumption that the flux curve in the slot is parallel to the slot bottom should be added. For semiclosed

slots shown in Fig. 10(a), the conventional analytical method evaluates the slot leakage specific permeance as

$$\lambda_{ss} = \frac{H_{s2}}{3B_s} + \frac{2H_{s1}}{B_{s0} + B_s} + \frac{H_{s0}}{B_{s0}}. \quad (22)$$

However, such assumption is reasonable for open slots but not for semiclosed ones, where the leakage flux distribution is shown in Fig. 9. It is obvious that a large proportion of flux curve is arched and not parallel to the slot bottom. Hence, the conventional assumption will cause inevitable error.

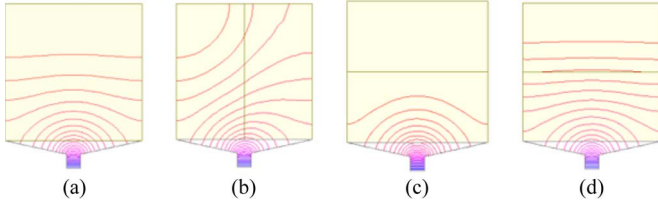


Fig. 9. Leakage flux distribution in the semiclosed slot. (a) Single layer. (b) Double layer (all teeth wound) with one layer excited. (c) Double layer with the upper layer excited. (d) Double layer with the lower layer excited.

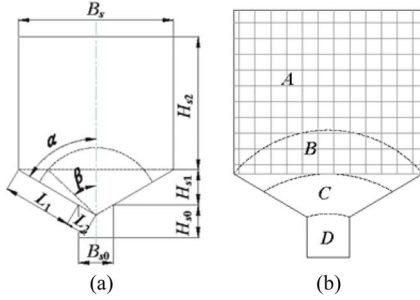


Fig. 10. Dimensions and region division of the single-layer winding slot. (a) Slot dimensions. (b) Region division.

1) *Improved Method for Single-Layer Windings:* For semiclosed slots with dimensions shown in Fig. 10(a), this paper separates the slot area into four regions according to the real leakage flux distribution shown in Fig. 9(a). In region A, the flux curve is regarded as linear and coupled with part of the conductors. In region B, the flux curve is arched and coupled with part of the conductors. In region C, the flux curve is also arched but coupled with all of the conductors. In region D, which is the slot opening, the flux curve is parallel to the slot bottom and coupled with all of the conductors.

Several parameters are defined in advance for the convenience of calculation.

Hypotenuse length of wedge

$$L_1 = \sqrt{(B_s - B_{s0})^2/4 + H_{s1}^2}. \quad (23)$$

Half of the wedge flare angle

$$\alpha = \arcsin\left(\frac{B_s - B_{s0}}{2L_1}\right). \quad (24)$$

Length of the wedge extended line

$$L_2 = \frac{B_{s0}/2}{\sin \alpha} = \frac{B_{s0}L_1}{B_s - B_{s0}}. \quad (25)$$

Maximum radius of region B

$$R_{\max} = \min(L_1 + L_2, L_2 \cos \alpha + H_{s1} + H_{s2}). \quad (26)$$

Half of the flare angle of region B

$$\beta_{\max} = \arccos\left(\frac{L_2 \cos \alpha + H_{s1}}{R_{\max}}\right). \quad (27)$$

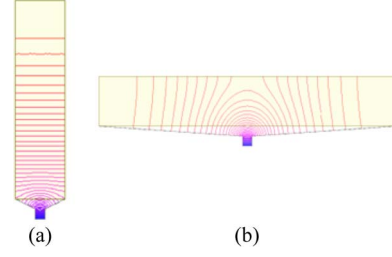


Fig. 11. Flux distribution in region A with various shapes. (a) Slim shape. (b) Flat shape.

In particular, if $L_2 \cos \alpha + H_{s1} + H_{s2} > L_1 + L_2$, $\beta_{\max} = \alpha$.
Area of region A

$$S_A = B_s H_{s2} - R_{\max}^2 \left(\beta_{\max} - \frac{1}{2} \sin 2\beta_{\max} \right). \quad (28)$$

The ratio of S_A to the area of the conductor region is defined as ζ

$$\zeta = \frac{S_A}{B_s H_{s2}} = 1 - \frac{R_{\max}^2}{B_s H_{s2}} \left(\beta_{\max} - \frac{1}{2} \sin 2\beta_{\max} \right). \quad (29)$$

The leakage specific permeance of each region is calculated in the following with prior assumptions made as follows.

- 1) The conductors and current distribute evenly in the conductor region of the slot.
- 2) The permeability of the stator core is large enough so that the magnetic curve cuts through slot edges perpendicularly.
- 3) All insulation is ignored.

a) *Leakage specific permeance of region A:* If region A is of slim shape, the entire flux curve can be regarded as parallel to the slot bottom, as shown in Fig. 11(a). Reforming region A to a rectangle with the same area and width, the equivalent height is then

$$H_A = S_A/B_s. \quad (30)$$

Moreover, the slot specific permeance is then

$$\lambda_{s11} = \frac{H_A}{3B_s} = \frac{\zeta B_s H_{s2}/B_s}{3B_s} = \frac{H_{s2}}{3B_s} \zeta. \quad (31)$$

However, if $H_{s2} \ll B_s$, the entire flux curve can be regarded as vertical to the slot bottom, as shown in Fig. 11(b). Reforming region A to a rectangle with the same area and height, the equivalent width is then

$$B_A = S_A/H_{s2}. \quad (32)$$

Then, the leakage specific permeance of region A is

$$\lambda_{s12} = \frac{B_A}{3H_{s2}} = \frac{\zeta B_s H_{s2}/H_{s2}}{3H_{s2}} = \frac{B_s}{3H_{s2}} \zeta. \quad (33)$$

In fact, the flux curve in region A is a combination of the two aforementioned cases, and the proportion depends on the ratio of B_s to H_{s2} . If B_s/H_{s2} is small, the proportion of λ_{s11} is also small and vice versa. Hence, the overall leakage

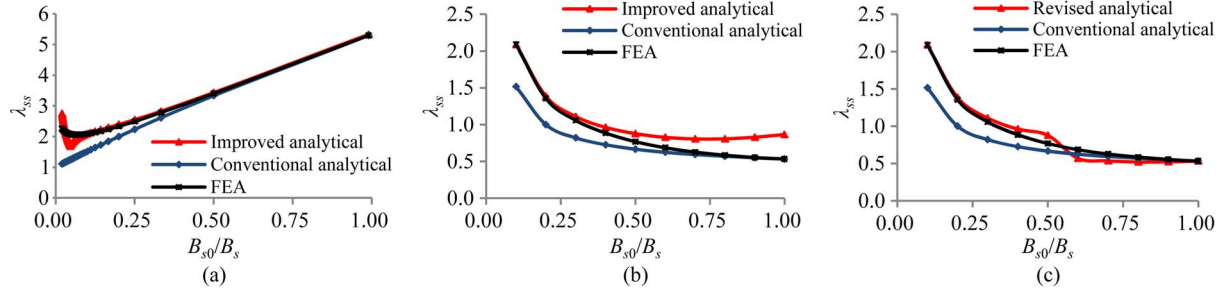


Fig. 12. Slot leakage specific permeance with respect to B_{s0}/B_s . (a) $B_{s0} = 1$ mm, while B_s varies. (b) $B_s = 10$ mm, while B_{s0} varies. (c) Revised work of (b).

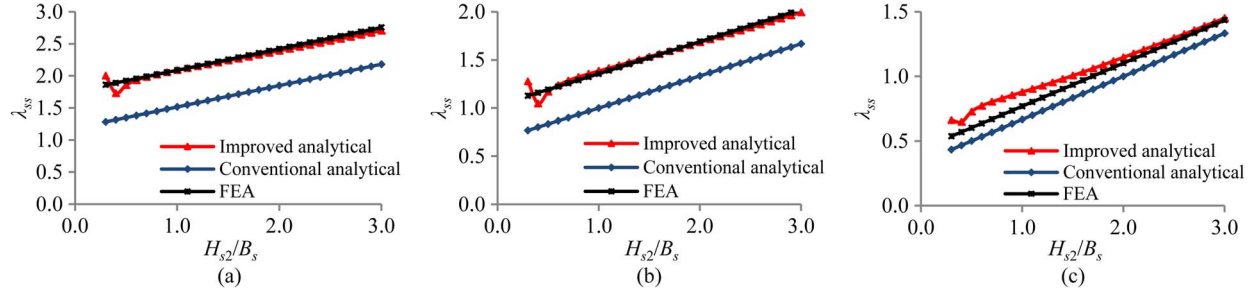


Fig. 13. Slot leakage specific permeance with respect to H_{s2}/B_s at different B_{s0}/B_s values. (a) $B_{s0}/B_s = 0.1$. (b) $B_{s0}/B_s = 0.2$. (c) $B_{s0}/B_s = 0.5$.

specific permeance of region A can be written as (34), which can balance the proportion to some extent by itself

$$\lambda_{s1} = \lambda_{s11} + \lambda_{s12} = \left(\frac{B_s}{3H_{s2}} + \frac{H_{s2}}{3B_s} \right) \zeta. \quad (34)$$

b) *Leakage specific permeance of region B:* The area of region B is

$$S_B = \alpha R_{\max}^2 - \alpha (R_{\max} \cos \beta_{\max})^2 \quad (35)$$

with the number of conductors

$$N_B = (1 - \zeta) N_s. \quad (36)$$

Take region B as a rectangle with the same area and N_B conductors distributed evenly. Let the equivalent height

$$H_B = R_{\max} - R_{\max} \cos \beta_{\max}. \quad (37)$$

The equivalent width

$$B_B = \alpha (R_{\max} + R_{\max} \cos \beta_{\max}). \quad (38)$$

Then, the leakage specific permeance of the equivalent region B is

$$\lambda_{s2} = \int_0^{H_B} \left[\zeta + (1 - \zeta) \frac{r}{H_B} \right]^2 \frac{dr}{B_B} = \frac{1 + \zeta + \zeta^2}{3\alpha} \tan^2 \frac{\beta_{\max}}{2}. \quad (39)$$

c) *Leakage specific permeance of region C:* For the fan-shaped region C without any conductors, its leakage specific permeance can be calculated as

$$\lambda_{s3} = \frac{1}{\alpha} \frac{B_s H_{s1} - B_{s0} L_1}{B_s H_{s1} + B_{s0} L_1}. \quad (40)$$

d) *Leakage specific permeance of region D:* The leakage specific permeance of the slot opening is

$$\lambda_{s4} = H_{s0}/B_{s0}. \quad (41)$$

Hence, the slot leakage specific permeance for the single-layer windings can be obtained by adding (34) and (39)–(41) as

$$\lambda_{ss} = \lambda_{s1} + \lambda_{s2} + \lambda_{s3} + \lambda_{s4}. \quad (42)$$

The proposed method is compared with the conventional and FEA, as shown in Figs. 12 and 13.

Fig. 12 illustrates the relationship between the slot leakage specific permeance and the ratio of B_{s0} to B_s , where B_s and B_{s0} vary, respectively. Other dimensions are kept constant, i.e., $H_{s0} = 1$ mm, $H_{s1} = 1$ mm, and $H_{s2} = 10$ mm. It can be seen that, when $B_{s0}/B_s = 1$, which represents the open slot case, the conventional method is accurate. However, when $B_{s0}/B_s < 1$, the error between the conventional method and FEA increases. In contrast, the improved method proves concordant with FEA and shares the same sunken trend at small B_{s0}/B_s , as shown in Fig. 12(a).

However, when the shape of the slot is close to a square as shown in Fig. 12(b), with the slot dimension of 10 mm $\times$ 12 mm when $B_{s0}/B_s = 1$, the improved method loses out to the conventional. This is because, when the conductor region is square and the slot opening is very large, λ_{s12} should be small, which is not reflected in (34). The revised results with λ_{s12} omitted are illustrated in Fig. 12(c), and it can be seen that the revised expression is also applicable to square-shaped open slots.

Fig. 13 illustrates the relationship between the slot leakage specific permeance and the ratio of H_{s2} to B_s at different B_{s0} values, during which $H_{s0} = 1$ mm, $H_{s1} = 1$ mm, and

$B_s = 10$ mm. It can be seen that the improved method coincides well with FEA, especially when B_{s0} is far smaller than B_s .

Therefore, the slot leakage inductance per phase for a single air gap can be obtained from

$$L_s = 2\mu_0 \frac{N^2}{pq} l_{\text{ef}} \lambda_s \quad (43)$$

where λ_s is the slot leakage specific permeance per phase for a single air gap ($\lambda_s = \lambda_{ss}$).

2) *Slot Leakage Inductance for Double-Layer Windings:* If all of the conductors in the slot belong to the same phase, the slot leakage inductance of double-layer windings is equal to that of single-layer ones. However, the more general case will be discussed in the following.

Let t be the greatest common divisor of slot number Q and pole number $2p$; then, the slot number in the unit machine is $Q_0 = Q/t$, and the pole number is $2p_0 = 2p/t$.

For balanced three-phase windings, there are $Q_0/3$ slots in the unit machine belonging to phase a . Assume that there are x slots in which both layers belong to phase a ; then, there are $Q_0/3 - x$ slots in which one layer belongs to phase b and the same to phase c . Hence, the total slot leakage flux linkage of phase a in unit machine is

$$\begin{aligned} \dot{\Psi} &= \sqrt{2}\mu_0 \left(\frac{N_s}{2}\right)^2 l_{\text{ef}} \left[(\lambda_1 \dot{I}_a + \lambda_{12} \dot{I}_a)x + (\lambda_2 \dot{I}_a + \lambda_{21} \dot{I}_a)x \right. \\ &\quad \left. + (\lambda_1 \dot{I}_a + \lambda_{12} \dot{I}_b)(Q_0/3 - x) \right. \\ &\quad \left. + (\lambda_2 \dot{I}_a + \lambda_{21} \dot{I}_c)(Q_0/3 - x) \right] \\ &= \sqrt{2}\mu_0 \left(\frac{N_s}{2}\right)^2 l_{\text{ef}} \\ &\quad \times \left\{ (\lambda_1 + \lambda_{12} + \lambda_2 + \lambda_{21})x \dot{I}_a \right. \\ &\quad \left. + [(\lambda_1 + \lambda_2)\dot{I}_a + \lambda_{21}\dot{I}_c + \lambda_{12}\dot{I}_b](Q_0/3 - x) \right\} \end{aligned} \quad (44)$$

where λ_1 and λ_2 are the self-leakage specific permeances of each layer and λ_{12} and λ_{21} are the mutual leakage specific permeances between two layers (generally, $\lambda_{12} = \lambda_{21}$).

Moreover, it is obvious that, if both layers belong to the same phase, the slot leakage flux linkage is equal to that of single-layer windings with the same exciting current, i.e.,

$$\mu_0 (N_s/2)^2 l_{\text{ef}} (\lambda_1 + \lambda_2 + \lambda_{12} + \lambda_{21}) = \mu_0 N_s^2 l_{\text{ef}} \lambda_{ss}. \quad (45)$$

Hence

$$\lambda_1 + \lambda_2 + \lambda_{12} + \lambda_{21} = 4\lambda_{ss}. \quad (46)$$

Considering

$$\dot{I}_b = \dot{I}_a e^{-j\frac{2\pi}{3}} \quad \dot{I}_c = \dot{I}_a e^{j\frac{2\pi}{3}} \quad (47)$$

and (46), the total leakage flux linkage of phase a in unit machine is obtained as

$$\dot{\Psi} = \sqrt{2}\dot{I}_a \mu_0 \left(\frac{N_s}{2}\right)^2 l_{\text{ef}} \left[4\lambda_{ss} \frac{Q_0}{3} - \lambda_{12} \left(\frac{Q_0}{3} - x\right) \right]. \quad (48)$$

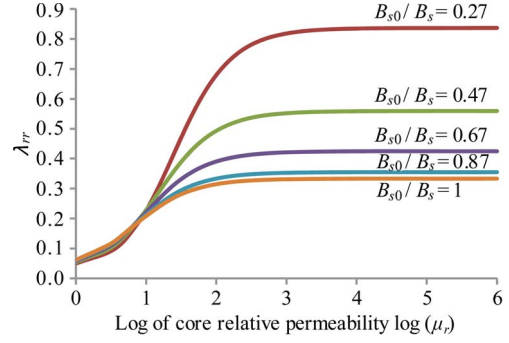


Fig. 14. Single-layer slot leakage specific permeance λ_{ss} with respect to core relative permeability at different B_{s0}/B_s values.

The slot leakage inductance per phase for double-layer windings is then

$$L_s = \frac{t\Psi}{\sqrt{2}I_a} \frac{1}{a^2} = 2\mu_0 \frac{N^2}{pq} l_{\text{ef}} \lambda_s \quad (49)$$

where the slot leakage specific permeance per phase is

$$\lambda_s = \lambda_{ss} - \lambda_{12} \cdot (1 - 3x/Q_0)/4 = \lambda_{ss} - k_\beta \lambda_{12} \quad (50)$$

and λ_{ss} is the slot leakage specific permeance for single-layer winding. k_β is a periodic function of β with the period of two and symmetrical in each period, so $\beta \in [0, 1]$ is enough in obtaining every k_β

$$k_\beta = \begin{cases} (4 - 3\beta)/4 & 0 \leq \beta \leq 1/3 \\ (5 - 6\beta)/4 & 1/3 < \beta \leq 2/3 \\ (3 - 3\beta)/4 & 2/3 < \beta \leq 1. \end{cases} \quad (51)$$

Hence, only the mutual leakage specific permeance between two layers, i.e., λ_{12} or λ_{21} , and the equivalent single-layer slot leakage specific permeance λ_{ss} are needed to obtain the slot leakage specific permeance for double-layer windings. Usually, the sum of λ_{12} and λ_{21} is close to that of λ_1 and λ_2 , i.e., $\lambda_{12} + \lambda_{21} = 2\lambda_{12} \approx \lambda_1 + \lambda_2$. Therefore, $\lambda_{12} \approx \lambda_{ss}$, and for double-layer windings, the slot leakage specific permeance per phase for a single air gap can be approximated as

$$\lambda_s \approx \lambda_{ss}(1 - k_\beta). \quad (52)$$

In particular, for full-pitch windings, $\lambda_s = \lambda_{ss}$.

3) *Effect of Core Saturation:* In fact, the permeability of the core cannot be as large as the assumption, and the slot leakage specific permeance will be affected by saturation. Fig. 14 shows the relationship between the slot leakage specific permeance of single-layer windings and the relative permeability of the core at different B_{s0}/B_s values. It can be seen that the slot leakage inductance has a negative relationship with saturation.

It should be noted that the effective air gap length also affects the slot leakage in practical machines. However, for AFPMS or SPMs which share large effective air gaps, the influence can be ignored.

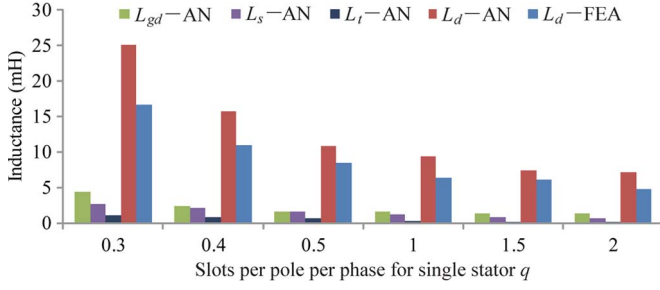


Fig. 15. Component and d -axis inductances of AFPMs with respect to SPP obtained from both analytical (AN) and FEA evaluations when the pole number remains invariant.

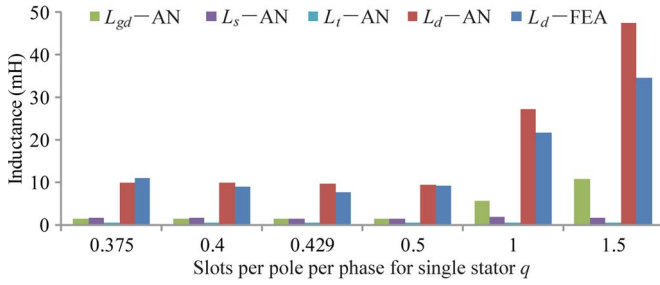


Fig. 16. Component and d -axis inductances of AFPMs with respect to SPP obtained from both analytical (AN) and FEA evaluations when the slot number remains invariant.

C. Tooth Tip Leakage Inductance

Because of the large air gap of AFPMs, part of the armature magnetic field leaks in the gap and causes tooth tip leakage inductance

$$L_t = 2\mu_0 \frac{N_1^2}{pq} l_{ef} \lambda_t \quad (53)$$

where λ_t is the tooth tip leakage specific permeance

$$\lambda_t = \frac{5\delta_1}{B_{s0}} \left/ \left(5 + \frac{4\delta_1}{B_{s0}} \right) \right. \quad (54)$$

V. VALIDATIONS OF VARIOUS SLOT-POLE COMBINATIONS

The d - and q -axis inductances of AFPMs can finally be obtained by substituting the component inductances deduced previously in (2). The proposed investigations are validated via FEA, as shown in Figs. 15 and 16. The physical parameters of these AFPMs are listed in Table II. For prototypes with the same number of poles, slot width B_s is determined to obtain similar filling factors. As q -axis inductances are quite close to d -axis inductances in these machines and less important during flux-weakening operations, only the latter is illustrated. To reveal the nature of the d - and q -axis inductances influenced by physical dimensions and slot-pole combinations, the permeability of the cores is set large enough so that it will be identical with the analytical calculations. All of the coil pitches are determined to obtain the largest fundamental winding factors.

Fig. 15 shows the relationship between d -axis inductance and SPP with the same pole number of 10, which is consistent with

TABLE II
PHYSICAL PARAMETERS OF AFPMS

Parameter	Same pole number	Same slot number
Machine Type	AFIR	
Winding Type	Double-Layer	
D_o (mm)	210	350
D_i (mm)	140	210
δ_1 (mm)	1	
h_m (mm)	5	
B_{s0} (mm)	1	2
B_s (mm)	-	15
$H_{s0}/H_{s1}/H_{s2}$ (mm)	1/1/9	
N_l	180	228
χ_{gd} (mH)	36.1~36.3	185.2

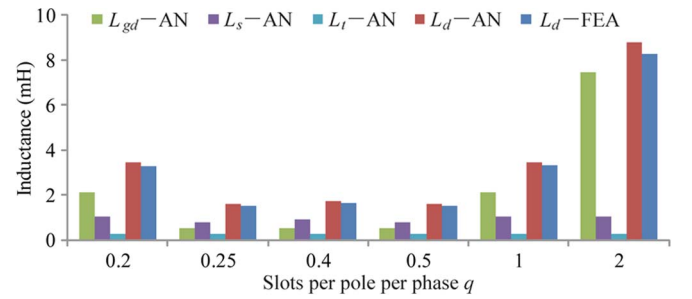


Fig. 17. Component and d -axis inductances of 12-slot RFPMs with respect to SPP obtained from both analytical (AN) and FEA evaluations.

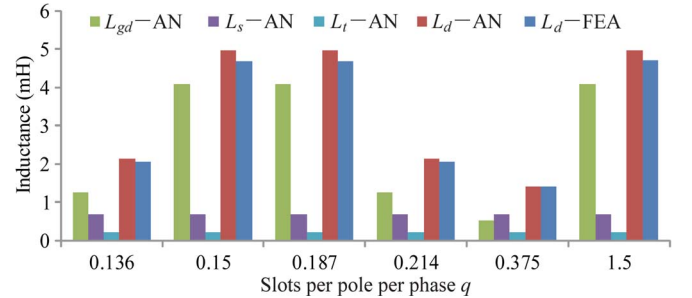


Fig. 18. Component and d -axis inductances of nine-slot RFPMs with respect to SPP obtained from both analytical (AN) and FEA evaluations.

El-Refai's discovery that fractional slot concentrated winding machines shared larger d -axis inductances because of larger air gap inductances. However, Fig. 16 shows the relationship with the same slot number of 36, from which the conclusion in Section IV-A is demonstrated that the d -axis inductances of fractional slot concentrated winding machines are not always larger than that of distributed ones with the same physical dimensions, i.e., $\chi_{gd,aq}$.

The investigations are also applicable to RFPMs, where the d - and q -axis inductances can be derived as

$$L_{d,q} = L_{gd,gq} + L_s + L_t + L_e \quad (55)$$

The 12-slot and 9-slot surface-mounted RFPMs with different pole numbers are evaluated with the proposed method and FEA, as shown in Figs. 17 and 18, respectively. The physical parameters of these prototypes are listed in Table III. The

TABLE III
PHYSICAL PARAMETERS OF RFPMS

Parameter	12 slots	9 slots
Machine Type	SPM	
Winding Type	Double-Layer	
Stator inner diameter (mm)	60	
Axial length (mm)	70	
δ_1 (mm)	1	
h_m (mm)	3	
B_{s0} (mm)	1	
B_s (mm)	10	15
$H_{s0}/H_{s1}/H_{s2}$ (mm)	1/1/10	
N_l	76	57
χ_{gd} (mH)	7.78	4.38



Fig. 19. Prototypes for tests. (a) Prototype I: 750-W servo motor with nine slots and eight poles. (b) Prototype II: 1-kW servo motor of 12 slots and 8 poles.

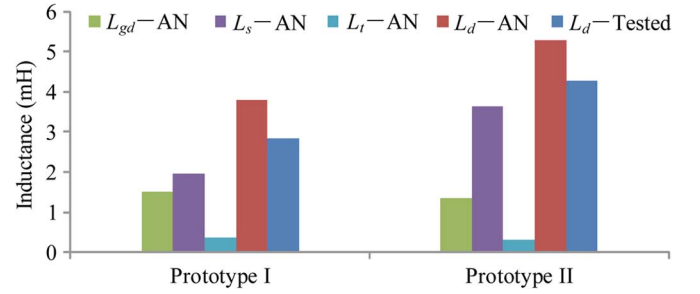
TABLE IV
PHYSICAL PARAMETERS OF THE PROTOTYPES

Parameter	Prototype I	Prototype II
Rated Power (W)	750	1000
Q/p	9/4	12/4
y_1	1	1
D_{avg} (mm)	36	61
l_{ef} (mm)	60	50
δ_1 (mm)	0.5	0.75
h_m (mm)	3	3.5
N_l	132	156
χ_{gd} (mH)	12.4	19.9

permeability of the cores is also set large enough so that it will be identical with the analytical calculations, and coil pitches are determined to obtain the largest fundamental winding factors. Similar conclusions can be obtained from these figures that distributed winding machines prevail when slot numbers are the same. Moreover, it can be seen that, when SPP is quite small, the d -axis inductances increase reversely because the windings are no longer concentrated but distributed in order to obtain the largest fundamental winding factors, showing a periodic law between d -axis inductance and SPP which is coherent with Table I.

VI. EXPERIMENTAL RESULTS

The d -axis inductances of two prototypes as shown in Fig. 19 with different physical structures and slot-pole combinations are tested for further verification of the proposed method. Their physical parameters are listed in Table IV. The inductance data

Fig. 20. Component and d -axis inductances of the two prototypes obtained from analytical (AN) evaluations and tests.

obtained from analytical evaluations and tests are illustrated in Fig. 20, which shows good concordance. It should be noted that, as the degree of saturation is very hard to estimate accurately, all of the analytical evaluations are based on the assumption that $K_{sd} = 1$, i.e., no saturation in cores.

VII. CONCLUSION

In this paper, a novel perspective of how d - and q -axis inductances are influenced by physical dimensions and slot-pole combinations has been proposed and applied to both AFPMs and RFPMS. The sum of the armature and differential leakage inductances was defined as the air gap inductance and was expressed as a product of Σg and $\chi_{gd,gq}$ so that the “inductance ability” of different slot-pole combinations can be quantified and can be independent of machine types and physical dimensions. Numerous Σg values of different slot-pole combinations were provided in order to help designers make proper choices in the view of inductance. With the help of Σg , the competition between fractional slot concentrated winding machines and distributed winding machines in d - and q -axis inductances was proved to be not one sided.

The d - and q -axis inductance expression for multigap AFPMs was proposed first, and the equivalent diameter for calculation was determined from the comprehensive observation of 3-D flux distribution. Then, the d - and q -axis component inductances were investigated, during which the improved method of calculating slot leakage inductance of semiclosed slots was also proposed and proved more accurate than the conventional method. Finally, calculations of both AFPM and RFPMS d -axis inductances with different slot-pole combinations were evaluated with the proposed method and validated via FEA, from which it could be seen that concentrated winding machines took advantage in d -axis inductances when pole numbers were the same but lost out to distributed ones when slot numbers were the same.

APPENDIX

The calculation of Σg is introduced here for three-phase windings with a phase belt of 60° . The following procedure is universal for both fractional and integral slot windings.

Let

$$\begin{cases} b = Q / \gcd(Q, 2mp) \\ d = 2mp / \gcd(Q, 2mp) \end{cases} \quad (\text{A.1})$$

where $\gcd(Q, 2mp)$ represents the greatest common divisor of slot number Q and the product of phase number m and pole number $2p$. Σg is calculated by considering the parity of d , respectively.

- 1) If d is odd, the orders of all of the harmonics generated by armature excitation satisfy

$$v_s = (6n + 1)/d \quad \text{or} \quad -(6n - 1)/d \quad (\text{A.2})$$

where $n = 0, 1, 2, \dots$

Find the minimum n_x to make $[6b(n_x - 1) + 1]/d$ an integer, where $n_x = 0, 1, 2, \dots$, and let

$$X = [6b(n_x - 1) + 1]/d. \quad (\text{A.3})$$

The coefficient of air gap specific permeance can be obtained

$$\Sigma g = \sum_{n=0,1,2,\dots} \left[\frac{1}{pv_s} \cdot \sin\left(v_s \frac{\pi\beta}{2}\right) \cdot \frac{\sin\left(\frac{\pi}{6} v_s \cdot d \cdot X\right)}{b \cdot \sin\left(\frac{\pi}{6} \frac{v_s \cdot d \cdot X}{b}\right)} \right]^2. \quad (\text{A.4})$$

- 2) If d is even, the orders of all of the harmonics generated by armature excitation satisfy

$$v_s = 2(3n + 1)/d \quad \text{or} \quad -2(3n - 1)/d \quad (\text{A.5})$$

where $n = 0, 1, 2, \dots$

Find the minimum n_x to make $[3b(n_x - 1) + 1]/d$ an integer, where $n_x = 0, 1, 2, \dots$, and let

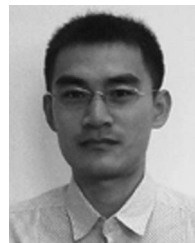
$$X = 2[3b(n_x - 1) + 1]/d. \quad (\text{A.6})$$

Then, the coefficient of air gap specific permeance can be calculated

$$\Sigma g = \sum_{n=0,1,2,\dots} \left[\frac{1}{pv_s} \cdot \sin\left(v_s \frac{\pi\beta}{2}\right) \cdot \frac{\cos\left(\frac{\pi}{12} v_s \cdot d \cdot X\right)}{b \cdot \cos\left(\frac{\pi}{12} \frac{v_s \cdot d \cdot X}{b}\right)} \right]^2. \quad (\text{A.7})$$

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